

A

Persona condition

Sample a profile source

Profile source bank

No profile



baseline

Empirical



graphics, survey

Synthetic



generated profiles

Game-grounded



task axes



sampled profile p

condition for the LLM

B

Matching assay

Identify with real human trajectories

Information about yourself

persona/profile text

Social interaction script

PGG transcript

with players

LLM matcher

Real candidate trajectories in the game

Player A

fully cooperative



contributions over rounds

Player B

conditional



pop. rate
no communication
contributions over rounds

Player C

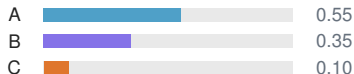
fully defective



cheap talk: "all in"
contributions over rounds

Matched distribution Q_g

probability mass over
real trajectories



C

Behavioral support

Compare matched and human distributions



human trajectories



matched mass

Empirical support P_g

uniform over observed

human trajectories

compare

Matched support Q_g

LLM-selected probability mass

ability mass

Diagnostics

Behavioral skew

$$\mathbb{E}_{Q_g}[f] - \mathbb{E}_{P_g}[f]$$

Behavior-space coverage

distance between P_g and

Residual identity concentration

collapse after grouping behaviorally similar players